

Low Inertia Power System Planning Considering Frequency Quality Under High Penetration of Renewable Energy

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Abstract—With the gradual replacement of synchronous generators by renewable energy sources, the power imbalance in the system is exacerbated by the fluctuations of renewable energy output. Simultaneously, traditional frequency regulation resources will become insufficient with the reduction of online capacity of thermal units. These trends both lead to a decrease in the performance of power system frequency regulation. The planning of power system generation, particularly frequency regulation resources, will significantly influence the future quality of frequency. This paper proposes a frequency-constrained generation planning method under a high-proportion renewable energy system. The performance of power system frequency regulation is analyzed based on the simulation of the load-frequency control dynamics, where both the frequency regulation from conventional generation and renewable energy are taken into consideration. Non-linear frequency constraints are extracted from the generated data set, and then embedded into the power system planning model by utilizing a sparse weighted oblique decision tree. The case study on HRP-38 system validates the effectiveness of the method. The influence of renewable energy frequency regulation on unit commitment is also investigated.

Index Terms—Low inertia, frequency regulation, frequency deviation, data-driven, power system planning.

NOMENCLATURE

A. Indices and Sets

w	Index of wind farms from 1 to W .
pv	Index of solar power stations from 1 to PV .

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g	Index of thermal units from 1 to G .
es	Index of energy storage stations from 1 to ES .
n	Index of bus nodes from 1 to N .
l	Index of transmission lines from 1 to L .
t	Index of time periods from 1 to T .
s	Index of scenarios from 1 to S .
c	Index of unit clusters from 1 to C .
Ω_n^W	Set of wind farms connected at bus n .
Ω_n^S	Set of solar power stations connected at bus n .
Ω_n^G	Set of thermal units connected at bus n .
Ω_n^{ES}	Set of energy storage stations connected at bus n .
$\Omega_n^{L(+)}$	Set of transmission lines starting from bus n .
$\Omega_n^{L(-)}$	Set of transmission lines ending at bus n .
Ω_c^G	Set of thermal units belonging to cluster c .
B. Parameters	
H^{sys}	Synthetic inertia of the system [s].
R^G	Droop coefficient of the thermal units [p.u.].
K	Renewable energy frequency regulation parameter [p.u.].
f_{db}	Frequency regulation dead band [Hz].
SC^{SU}	Start-up cost of thermal units [\$/MW].
V^C	Variable operating cost [\$/MWh].
$VoLL$	Cost of load shedding [\$/MWh].
$D_{n,t}^{Fore}$	Hourly forecasted load demand [\$/MW].
$\lambda_{w,t}^{Fore}$	Hourly forecasted wind generation [\$/MW].
$\lambda_{w,t}^{Fore}$	Hourly forecasted wind generation [\$/p.u.].
$\lambda_{s,t}^{Fore}$	Hourly forecasted solar generation [\$/p.u.].
Cap	Installed capacity of power plants [MW].
Cap_p	Installed power capacity of energy storage [MWh].
Cap_e	Installed energy capacity of energy storage [MWh].
λ_g^{Min}	Minimum output level [\$/p.u.].
α_g^{Ru}	Hourly ramp up rate [p.u.].
α_g^{Rd}	Hourly ramp down rate [p.u.].
T_g^{on}	Minimum on time periods [hrs].
T_g^{off}	Minimum off time periods [hrs].
N_c^G	Number of units in unit cluster c .
F_c^{Max}	Transmission capacity limit of line l [MW].
B_l	Susceptance for line l [p.u.].
IC	Annualized capital cost [CNY/kW-yr].

C. Variables

$C_{sys}^{Operation}$	Overall system operational cost [\$].
$C_{g,t}^{SU}$	Start-up cost of thermal units [\$].
$P_{g,s,t}$	Power generation of a thermal unit g [MW].
$P_{c,s,t}$	Aggregated power output of the thermal units within cluster c [MW].
$P_{w,s,t}$	Power output of a wind farm w [MW].
$P_{pv,s,t}$	Power output of a solar power unit pv [MW].
$P_{es,s,t}^{ch}$	Power charging of a energy storage unit es [MW].
$P_{es,s,t}^{dis}$	Power discharging of a energy storage unit es [MW].
$F_{l,s,t}$	Power flow of line l [MW].
$\theta_{n,s,t}$	Power angle of bus node n .
$O_{c,s,t}$	Online capacity of unit cluster c [MW].
$SU_{c,s,t}$	Start-up capacity of unit cluster c [MW].
$SD_{c,s,t}$	Shut-down capacity of unit cluster c [MW].
X_g	Binary variables of generations investment..
X_w	Binary variables of wind farms investment..
X_{pv}	Binary variables of solar power stations investment.

I. INTRODUCTION

A. Motivation

WITH the increasing penetration of renewable energy sources, power systems have been witnessing an increased imbalance and a decrease in the inertia and frequency regulation capabilities of conventional generators. This could jeopardize the frequency control performance when the penetration of renewable energy exceeds a certain level.

Recent research has shown a decline in frequency quality globally, as evidenced by an increasing trend of “minutes outside the band” reported by ENTSO [1], and an increase in frequency variance in the British power grid from approximately 0.05 Hz to approximately 0.065 Hz, as shown in Fig. 1 [2].

To address this issue, additional flexible resources must be utilized for frequency regulation. Apart from conventional thermal or hydro generators, batteries are regarded as an ideal resource to support frequency regulation. Wind turbine and photovoltaic panel can also participate in inertia, primary and secondary frequency regulation under deloading operation state. In addition to this, the hybrid energy storage system (HESS) and supercapacitor can be applied to smooth the frequency deviation [3] [4]. Therefore, the adequacy of such flexible resources should be considered in power system planning with high penetration of renewable energy. In fact, the coordination of the flexible resources that participate in frequency regulation is a tech-economic coupled problem: 1) The technical characteristics (delay, ramp capacity) of various flexible resources are distinct and may have synergies [5]. 2) Power system frequency regulation requirements over varying hours/days/weeks would be diversified because of the intermittency of renewable energy. The cost-effectiveness of each flexible resource relies on the combination of the various scenarios over a long-term period. 3) Furthermore, the relationship between the available capacity of flexible resources and the frequency control performance is

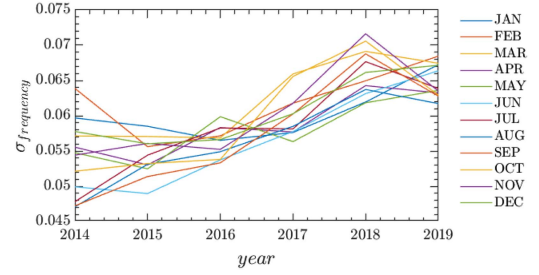


Fig. 1. Frequency variance of England from 2014 to 2019.

not a simple linear relationship. These issues lead to the problem that the frequency regulation capacity may not be adequate for all of the operation scenarios to meet the frequency control performance. The adequacy of flexible resources should be considered in power system planning problem.

B. Literature Review

The frequency quality of the power system includes two aspects: the frequency security after failures and the frequency fluctuations in the non-fault state. The frequency security denotes the ability to quickly adjust the active power output after the contingency, in order to prevent or reduce under frequency load shedding (UFLS). As opposed to frequency security after contingency, frequency control performance under normal operations concentrates on eliminating the long-term small power fluctuations, in order to avoid excessive wear and tear on the generators. Many researches have been conducted on maintaining the frequency quality in power system operation and planning. For power system frequency under contingency, research mainly focuses on typical indicators such as maximum RoCoF (Rate of Change of Frequency), the frequency nadir and a quasi-steady state frequency deviation. Some researches construct constraints of frequency security after contingency using model-driven or data-driven method. Model-driven methods employ a simulation based on the equivalent model in Laplace transform to estimate these indicators [6]. By solving the equivalent model, reference [7] reveals the functional relationship between the three indicators and magnitude of disturbance, system inertia, governor parameters, which are later formed as a set of linear constraints to formulate the feasible region of unit commitment. Reference [8] proposed an equivalent aggregated system frequency (ASFR) model and analyses the contribution of frequency support from renewable energies. Data-driven methods, such as decision tree, are also used to estimate and identify the frequency security region of power system in reference [9]. Rules from decision tree can guide the frequency modulation and load shedding modes of power system to improve the frequency security. Further, in reference [10], frequency constraints are generated by deep learning and encoded in the unit commitment.

Relatively few papers have focused on frequency regulation performance under non-fault conditions. Frequency control performance denotes the ability to maintain the system frequency deviation within a given limit in the long-term operation. Therefore, the frequency regulation performance is usually quantified

by the power frequency fluctuation and fluctuation of the tie line in multi-region power system. Based on this, different countries and regions have certain standards for frequency control performance, such as the BAAL [11] and CPS1 [12] standards of the North American power grid, the European power grid standards [13], and the CPS1 and CPS2 standards of China's power grid [14]. In [15], the authors explored the coupling relationship between primary frequency control and secondary frequency control (Automatic Generation Control, AGC). It was observed that the standard deviation of the frequency is profoundly impacted by the dead zone, aggregated droop of the primary frequency control, and the renewable penetration [16].

Only a limited number of pioneering studies have recognized the significance of incorporating frequency control performance into the realm of power system operation or planning. For example, in [17], the authors considered the influence of secondary frequency control by incorporating the operating cost of frequency control. Similarly, [18] embedded the secondary frequency control constraints into the market clearing process. In addition, [19] developed simple piecewise linear curves to represent the relationship between conventional generation and storage capacity, while [20] determined the size of the reserve capacity needed in the system based on forecasted power variation ranges from wind, solar, and load. However, these studies primarily focused on calculating the appropriate size of reserve capacity and did not thoroughly analyze the complex dynamic processes associated with frequency regulation that may arise under high penetration of renewable energy.

Some literature has employed dynamic process simulations to analyze the changes in frequency quality in power systems under high proportions of renewable energy, thereby highlighting the influence of renewables on frequency regulation. For instance, reference [14] revealed that a high penetration of wind power leads to fluctuations in the system frequency as the wind power capacity increases. Furthermore, a detailed analysis conducted on a small test system in reference [21] revealed frequency fluctuations that exceeded the permissible range when wind penetration exceeded 10%. Similarly, reference [22] investigated the frequency deviation of Thailand's power system in 2030, demonstrating that the maximum frequency deviation could exceed the standard with the increasing solar and wind power generation capacity, reaching up to 3,940 MW and 1,977 MW, respectively. Thus, ensuring the regulation performance of frequency becomes an inevitable concern in future-oriented power system planning problems. However, to the best of our knowledge, no research has thoroughly analyzed this issue.

It is important to note that the current simulation-based methods fall short in generating the necessary constraints for power system planning. Power system planning problems with frequency quality constraints often encounter computational challenges compared to the unit commitment and economic dispatch models. This is primarily due to the inclusion of unit investment scale considerations in addition to unit output, resulting in a higher dimensionality of variables. Additionally, the inclusion of frequency quality constraints introduces additional binary variables. For instance, the frequency constraints for contingency in [10] involve thousands of equations and hundreds of binary

variables for each single period. This significantly increases the computational burden in the unit combination problem. In the context of planning, the complexity of these constraints becomes even more challenging due to the increased dimensionality of the problem.

C. Contributions

In general, the current power system planning model cannot capture the complex relationship between the combination of secondary frequency regulation resources and the frequency regulation performance under high penetration of renewable energy. In this article, we formulate such a relationship in power system planning by using a machine learning method based on the simulation of load-frequency control dynamics. The contributions are threefold:

- We develop a frequency dynamic simulation model to analyze the system frequency regulation performance that considers both the conventional generator and renewable energy.
- The relationship between the power system generation mix and the system frequency regulation performance is built in a data-driven way based on simulation samples.
- A power system generation planning model is built that embeds the constraints for system frequency regulation performance based on the extracted rules.

D. Organization

The rest of the article is organized as follows. Section II presents the formulation of the problem and proposes the overall framework. Section III introduces the frequency regulation simulation model to generate sufficient samples. Sections IV and V extract the rules from the samples and embed them into power system planning models. Section VI presents a case study, and Section VII concludes.

II. PROBLEM STATEMENT AND FRAMEWORK

A. Frequency Regulation in Power System

The frequency control performance of the power grid depends on the real-time balance of active power in the system. Short time scale fluctuations caused by load and renewable energy is unavoidable, resulting in the system's continuous power imbalance. At this time, the rotor of the synchronous generator will release or absorb part of the kinetic energy to maintain the power balance in the system, leading to a decrease or increase of the system frequency.

The frequency deviation of the system can be obtained from the swing equation, which contains several variables, including system inertia constant H_{sys} , load-damping constant D , overall power imbalance in the system ΔP^{imb} and the frequency response of each unit in the system. Thermal units, wind farms, solar power stations and storages will be all required to provide frequency response in a high percentage renewable energy power system. The response of these units depends on the deviation of the system frequency, the AGC signal and the dynamic characteristic of each unit.

In this article the system frequency dynamic are modeled using the multi-machine system frequency response (MM-SFR) model, in which the dynamics of each unit is modeled separately using algebraic differential equations [23] (for details see Section III).

B. Vectors of Power System Operating States

The key to maintaining the frequency quality is to ensure that the frequency regulation resources available in each time period are sufficient to meet the power fluctuation regulation demand. This is especially acute in power systems with a high proportion of variable renewable energy (VRE). This is due to VRE being the main source of inevitable fluctuation. The demand for frequency regulation resources is variable in different time periods of the day. Furthermore, thermal power units with extremely low output or in shutdown are unable to provide regulation capacity and droop response. As a result, the frequency quality under a high proportion of renewable energy is heavily dependent on the power system's operating states.

Specifically, the frequency control performance is determined by a series of key operating factors: the amplitude of power imbalance imposed by load and renewable energy, available regulation capacity of conventional generators and renewable energy. Thus, we define the concept of power system operating state characterised by such operating parameters to study their relationship with frequency control performance. Mathematically, the operating state $x_{s,t}$ can be modeled using vectors:

$$x_{s,t} = [\sigma_{s,t}^{load}, \sigma_{s,t}^W, \sigma_{s,t}^{PV}, O_{1,s,t}, R_{1,s,t}^{up}, R_{1,s,t}^{down}, O_{2,s,t}, R_{2,s,t}^{up}, R_{2,s,t}^{down}, \dots, O_{c,s,t}, R_{c,s,t}^{up}, R_{c,s,t}^{down}, P_{s,t}^{Reg}] \quad (1)$$

where $x_{s,t}$ denotes the operating state, including online capacity of unit, ramp up and down capacity of unit, fluctuation of load and renewable energy. Here, we make the following assumption that the control parameters and the dynamic characteristics of each unit such as dead band, control delay and filter parameters remain unchanged during the whole planning period.

C. Problem Statement

We find the relationship between the frequency regulation performance and operation state and then explore how the performance requirement affects the generation investment. The problem can be divided into three sub-problems:

- 1) Modeling the influence of renewable energy on frequency regulation performance through models.
- 2) Developing discriminative rules from data sets using a data-driven approach.
- 3) Incorporating the derived rules into the planning model.

D. Framework

Firstly, this article presents a frequency regulation simulation model for high proportion of renewable energy, incorporating the statistical distribution characteristics of new energy and load

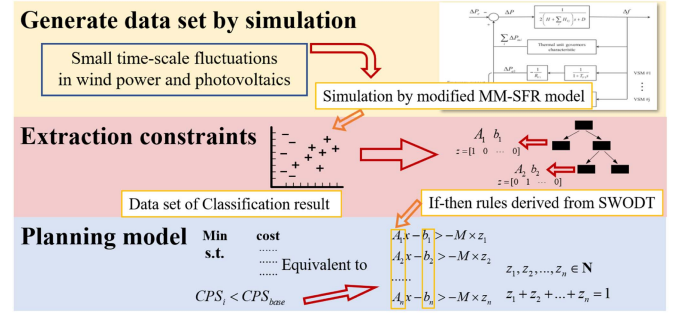


Fig. 2. Proposed framework for frequency performance constrained planning problem.

fluctuations, with consideration to the frequency response of renewable energy and addition of secondary frequency regulation to eliminate steady-state frequency deviation. Through analysis of a multitude of operation states in order to cover sufficient operation space, the frequency quality of a specific operating state can be determined. The technical details of the simulation are provided in Section III.

Secondly, the relationship between frequency regulation quality and system operation state is extracted based on the simulation data obtained by the first step. The frequency regulation performance requirement is transformed into the rule of the operation state of the power system by the sparse weighted oblique decision tree (SWODT) [9] learned from the simulation data. The details of the learning are described in Section IV.

Finally, the rules learned from the SWODT are extracted as mixed-integer linear constraints. The constraints are further embedded into a generation planning model. Then, the validity of the proposed method is verified on the HRP-38 system. The details are presented in Sections V and VI.

III. FREQUENCY REGULATION SIMULATION MODEL UNDER RENEWABLE ENERGY PENETRATION

The purpose of frequency regulation simulation is to evaluate whether frequency regulation performance of a specific operating state meets the system requirement, e.g., frequency standard deviation within ± 0.1 Hz. The simulation employs a modified multi-machine system frequency regulation (MM-SFR) model considering secondary frequency regulation. Such simulation generates a series of labeled datasets that will be later used to extract the boundaries of frequency performance requirements.

The overall frequency regulation simulation model is built, as shown in Fig. 3. The power system's overall imbalance considers both the fluctuation of renewable energy and load. Primary and secondary frequency responses from both thermal units and renewables are considered.

A. Modeling of Traditional Frequency Regulation Resource

Traditional frequency regulation resource is shown in Fig. 3, which includes primary frequency regulation and AGC with ramp limit and saturation. The primary frequency regulation is directly responded by the controller with a constant droop

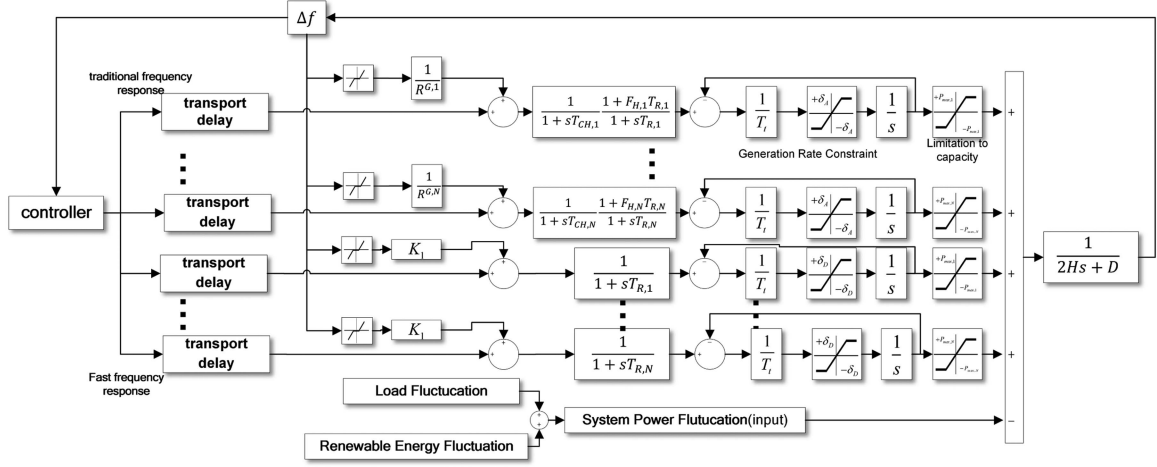


Fig. 3. Simulation model for frequency regulation performance under high-proportion renewable energy.

coefficient. Specifically, when a thermal power unit detects a frequency deviation, a signal of the output power change is added to the generator:

$$\Delta P^{g,PFR} = \begin{cases} -\frac{1}{R^g}(\Delta f + f_{db}) & \Delta f \leq -f_{db} \\ 0 & -f_{db} < \Delta f \leq f_{db} \\ -\frac{1}{R^g}(\Delta f - f_{db}) & \Delta f > f_{db} \end{cases} \quad (2)$$

where $\Delta P^{g,PFR}$ denotes the primary response of generator, f_{db} denotes the dead zone. In this article, $f_{db} = 0.033$ Hz.

In addition, the unit receives AGC signal, $\Delta P^{g,SFR}$, from the controller. The primary frequency regulation and the secondary frequency regulation do not affect each other according to the regulation. The actual output adjustment of the unit is written as (3) considering dynamics:

$$\Delta P^g = \frac{1}{1 + sT_{CH}} \frac{1 + sF_H T_R}{1 + sT_R} (\Delta P^{g,PFR} + \Delta P^{g,SFR}) \quad (3)$$

where the three equivalent parameters (T_{CH} , T_R and F_H) represent the combined effect of turbine governors. The parameters of the thermal power unit can be found in [8]. The generator rate constraint (GRC) for the traditional frequency regulation is set as 10% p.u./min. The settings is from [24].

B. Modeling of the Dynamic Frequency Regulation Resource

Renewable energy and electrochemical storage can also provide primary and secondary frequency regulation, which are regarded as dynamic frequency regulation resource. For renewable energy, it requires new control strategies such as deloading control, which allows renewable energies operate under maximum power point and leave a backup to support system frequency regulation. Fig. 4 shows how this control mode works. For this article, we assume that the renewable energy can be modified in terms of hardware and control logic to allow up to 10% of its capacity to be left as a backup. Additionally, we assume that downward regulation capacity is equal to upward regulation capacity, as is commonly seen in regulation of electricity markets.

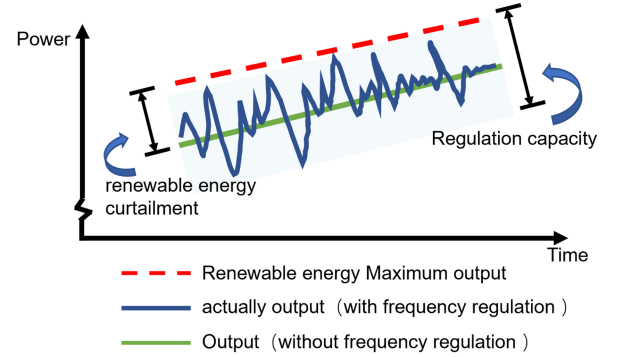


Fig. 4. Schematic diagram of frequency regulation of renewable energy through deloading mode.

Primary frequency regulation provided by renewable energy and electrochemical energy storage are:

$$\Delta P^{R,PFR} = \begin{cases} -K(\Delta f + f_{db}) & \Delta f \leq -f_{db} \\ 0 & -f_{db} < \Delta f \leq f_{db} \\ -K(\Delta f - f_{db}) & \Delta f > f_{db} \end{cases} \quad (4)$$

where $K\Delta f$ represents droop control. K is frequency regulation parameters. In this article, K is set to be 15 [8]. f_{db} denotes the dead zone. In this article, $f_{db} = 0.033$ Hz.

Similar to the traditional resources, dynamic resources also receive AGC signals, $\Delta P^{R,SFR}$:

$$\Delta P^R = \frac{1}{1 + sT_V} (\Delta P^{R,PFR} + \Delta P^{R,SFR}) \quad (5)$$

where T_V represents the time constant of renewable energy. Compared with thermal power, the frequency adjustment of dynamic resources is faster and more suitable for responding to rapid fluctuations in the system. The GRC for the dynamic frequency regulation is set as 300%p.u./min.

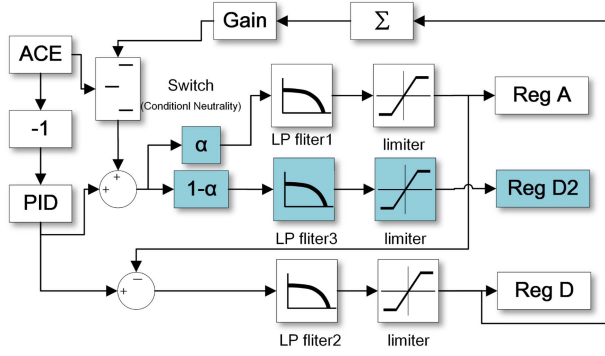


Fig. 5. AGC signal controller (modification is marked in blue).

C. Modeling of AGC Controller

In this article, the AGC signal is generated by a controller that is designed based on the PJM white paper. This controller manages rapid and slow fluctuations of power imbalance. It comprises two signals, Traditional Regulation (Reg A) and Dynamic Regulation (Reg D), which are divided by two low-pass filters. Reg D is sent to dynamic frequency regulation resources and designed such that the energy offset of these resources following Reg D signal can be compensated for by Reg A. This mechanism is referred to as “Conditional Neutrality” and is enabled by a block “switch”. Further details on the control, including parameters of the PID and the two low-pass filters, may be found in reference [25].

To adapt to the power system with a high proportion of renewable energy, some modifications are made to the controller, which are marked with blue in Fig. 5. Considering that deloading control of renewable energy does not have an energy offset problem, we generate an extra Reg D2 signal that does not consider energy offset. The signal is generated by adding a split before Filter 1 by using a coefficient α :

$$\alpha = \frac{Cap_{RegA} + Cap_{RegD}}{Cap_{RegA} + Cap_{RegD} + Cap_{RegD2}} \quad (6)$$

where Cap_{RegA} , Cap_{RegD} and Cap_{RegD2} represent the total regulation capacity of traditional regulation resources, electrochemical storage and renewable energy respectively. α denotes the proportion of Reg D signal (Dynamic frequency regulation resources except renewable energy) and Reg A signal over the regulation capacity of all resources. Low-pass filter 3 is identical to low-pass filter 2. The accumulation of the Reg D2 is not measured to compensate for the energy offset problem so that it responds to some slow fluctuations. In this way, renewable energy regulation that responds to Reg D2 has an effect of substituting some of the Reg A resources. The purpose of such change is to make the split of the signal more suitable for renewable energy to participate in frequency regulation.

After signal generation, there is a nominal network transport delay as data is sent to the plant for consumption. The transport delay is set to 8 seconds in this paper [25].

D. Generating Second Level Load and Renewable Energy Variation

The variations in the load and renewable energy sources are reflected on both the second and hour level. Frequency regulation is used to counteract the energy imbalance from the second-level variations, while the hour-level variations are addressed through economic dispatch.

In this article, the second-level load and renewable energy variation is generated by a stochastic differential equation technique, which is capable of accurately capturing both the probability distribution and the intensity of the original curve [21].

Mathematically, the renewable energy and load P_t are generated by (7):

$$dP_t = -\theta(P_t - \mu)dt - \sigma dW_t, t > 0 \quad (7)$$

where dP_t is the differential of P_t , σ is the variance of the normal distribution. θ is autocorrelation coefficient of the variation. W_t is a standard Brownian process.

The parameters of the variation of load and renewable energy are extracted from their historical time-series measurements with one-second resolution. Their long-term variation, which is met by the economic dispatch of generators, is removed from the historical time series beforehand using the moving average method:

$$\begin{cases} P_t^{long} = \frac{P_{t-(\frac{N}{2}-1)} + P_{t-(\frac{N}{2}-2)} + \dots + P_t + \dots + P_{t+(\frac{N}{2})}}{N} \\ P_t^{short} = P_t - P_t^{long} \end{cases} \quad (8)$$

where P_t^{long} represents the long-term variation in the system, while P_t^{short} represents the short-term variation. N represents the length of time in the moving average, which is chosen as 5 minutes in this article.

E. Illustrated Sample of Simulation Results

The generated second level load and renewable energy variation curves are further used to evaluate frequency quality for each period. For each operation state, a separate time series with 50000 s is generated. Fig. 6 is part of the variation generated by (7).

Therefore, we generate fluctuation curves and evaluate frequency quality for each period.

The frequency regulation performance is evaluated through the simulation of a sufficient time step. In this article, the simulation is conducted for 50000 s with a step length of 0.01 s. σ_f is the variance of frequency, which is calculated based on the simulation results as the frequency regulation performance index.

$$\sigma_f = \sqrt{\frac{1}{N} \sum_{t=1,2,\dots,N} \Delta f_t^2} \quad (9)$$

where Δf_t is the difference between the actual frequency and the standard frequency. We use the above framework to assess the frequency control performance of the system for all operating states. We perform sampling on the key operating factors in (1) and obtain a dataset of operating states and their labels for whether the frequency regulation performance satisfies the

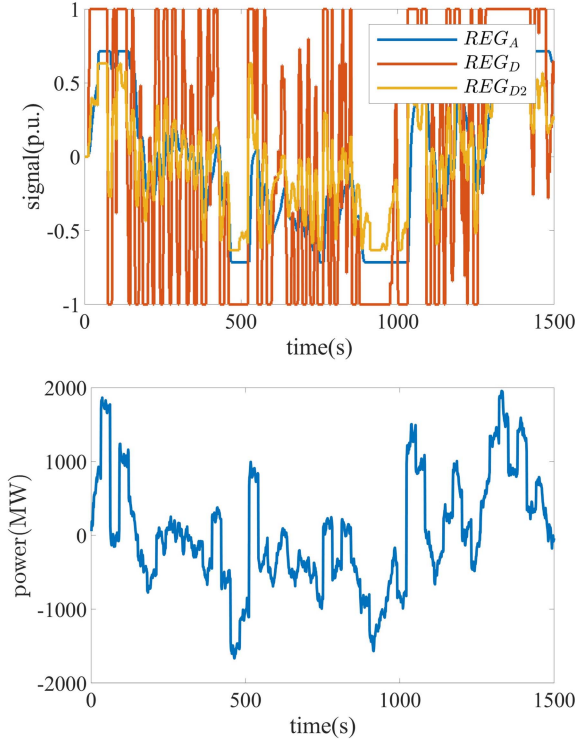


Fig. 6. AGC signal (above) and power variation curves (below).

requirements. The dataset is further used for the extraction of the rules in the next section.

IV. NUMERICAL EVALUATION AND RULE EXTRACTION

In this section, we use the results of direct time-domain numerical simulations to extract power system operation rules under certain frequency regulation performance, which are further embedded in the planning model. Extracting rules from labeled datasets is regarded as a classification problem and is handled by using a data-driven method.

SWODT is adopted for rule extraction. It is an improved version of the decision tree, where each internal node is a linear algebra inequality with respect to the system operation variables, i.e., the output of each generator, their frequency regulation parameter and the load on each load. Each leaf node tells whether the system frequency regulation meets the requirement (i.e., with variance less than 0.060 Hz) under the operation states described by the upstream inequalities.

Compared with those of the conventional supervised learning methods, the SWODT has two advantages: 1) Analytical linear rules. Compared with XGBoost, neural networks, SWODT is easier to embed into an optimization problem since it can be written as mixed-integer linear rules instead of nonlinear ones. 2) Less complexity. Compared with the classic decision tree, SWODT employs significantly fewer layers and leaves to reach the same accuracy because of its regularization during the learning process. It thus brings in much fewer binary variables when transformed into MILP constraints, which induces less time in solving the optimization problem.

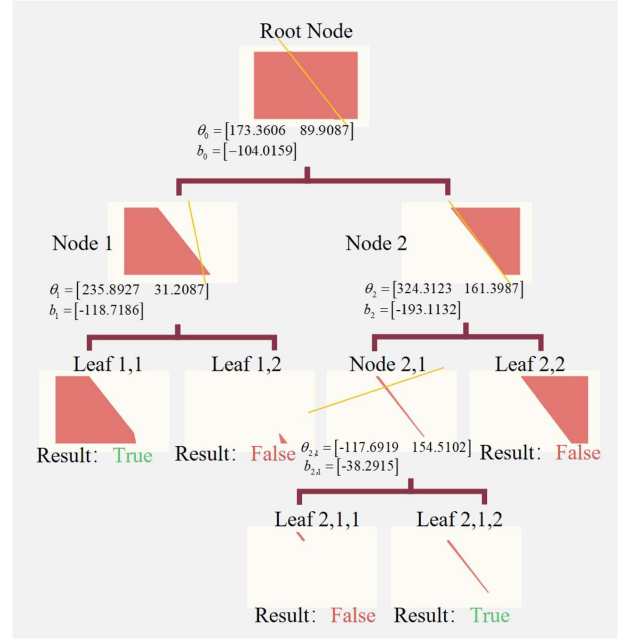
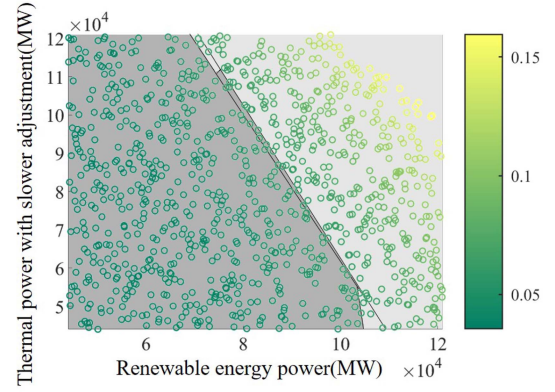


Fig. 7. The results and the decision vectors of each node of the decision tree.

Fig. 7 provides an illustration of the classification performance of SWODT in the HRP-38 test system. The rule is extracted by using the cases with various outputs of renewable energy and thermal power units, while other factors are unchanged. The decision tree uses merely three layers to demarcate the boundary in this case with an accuracy of 95.5%. The dark gray part indicates that the frequency regulation performance meets the standard ($\sigma_f < 0.060$ Hz), while the light gray part indicates that the frequency regulation performance does not. The learned decision tree is shown in Fig. 7, where the small subfigure on each node is its decision vector.

The decision tree model can directly determine whether the operation state $x_{s,t}$ satisfies the frequency quality requirements. For each leaf node k , we assume that its inequality can be written as $A_k x_{s,t} + b_k > 0$, where A_k and b_k are parameters of leaf node k . Its label for whether frequency quality requirements are met is presented by q_k :

$$q_k = \begin{cases} 1 & \text{if frequency requirements are met} \\ 0 & \text{if not met} \end{cases} \quad (10)$$

We also define $F_k(x_{s,t})$, which represents whether $x_{s,t}$ satisfies the inequality of the k st leaf:

$$F_k(x_{s,t}) = \begin{cases} 1 & \text{if } A_k x_{s,t} + b_k \geq 0 \\ 0 & \text{if } A_k x_{s,t} + b_k < 0 \end{cases} \quad (11)$$

Each operation state can only be classified into one leaf node of the decision tree, that is, $\sum_{c=1}^C F_c(x_{s,t}) = 1$.

We further define $G(x_{s,t})$ as the function of identifying whether the state $x_{s,t}$ meets the frequency requirement by using the learned SWODT model, mathematically:

$$G(x_{s,t}) = \sum_{k=1}^K q_k \times F_k(x_{s,t}) \quad (12)$$

In the planning optimization problem, the constraints that each operating state of the planning should meet the frequency quality requirements can be written as:

$$G(x_{s,t}) = 1 \quad \forall s, \forall t \quad (13)$$

Constraints such as (13) with function $F_c(x_{s,t})$ in the form of if-then cannot be directly embedded in the optimization problem, so the big-M method is adopted to convert the constraints into equivalent mixed integer constraints.

$$A_k x_{s,t} + b_k > -M \times (1 - z_{k,s,t}) \quad \forall k \in \Omega_{true}^{leaf}, \forall s, \forall t \quad (14)$$

$$\sum_{k \in \Omega_{true}^{leaf}} z_{k,s,t} = 1 \quad \forall s, \forall t \quad (15)$$

where Ω_{true}^{leaf} is the set of all the leaf nodes that satisfy $q_k = 1$, representing the operating states belonging to the node that meets the frequency quality requirements.

V. PLANNING MODEL CONSIDERING FREQUENCY REGULATION PERFORMANCE

This section integrates the frequency regulation performance requirement constraints into a generation planning model. We first introduce the traditional planning constraints and then describe how to link the variables of the traditional planning model with the decision tree constraints generated in Section IV. Finally, we list the mathematical forms of the traditional and modified planning models separately.

A. Traditional Generation Planning Constraints

1) Investment Constraints:

$$C^{investment} = \sum_{g=1}^G IC_g X_g + \sum_{w=1}^W IC_w X_w + \sum_{pv=1}^{PV} IC_{pv} X_{pv} \quad (16)$$

$$Cap_g = G_g^{cap} X_g; Cap_w = G_w^{cap} X_w; Cap_{pv} = G_{pv}^{cap} X_{pv}; \quad (17)$$

Equations (16) and (17) describe investment cost, including the cost of thermal units, wind farms and solar power stations. In this article, investment costs are expressed in terms of the equivalent annual cost of the unit, thus better reflecting the cost differences between the units.

2) Unit Commitment Constraints:

$$C^{operation} = \sum_{t=1}^T \sum_{g=1}^G C_{g,t}^{SU} + \sum_{t=1}^T \sum_{g=1}^G Vc_{g,t} P_{g,t} + \sum_{t=1}^T \sum_{n=1}^N VOLL \times VD_{n,t}^{cur} \quad (18)$$

Equations (18) describe operation cost, including the start-up cost, the variable cost and the load shedding cost over various scenarios and time periods.

$$\begin{aligned} & \sum_{g \in \Omega_n^G} P_{g,s,t} + \sum_{w \in \Omega_n^W} P_{w,s,t} + \sum_{pv \in \Omega_n^{PV}} P_{pv,s,t} \\ & + \sum_{es \in \Omega_n^E} (P_{es,s,t}^{dis} - P_{es,s,t}^{ch}) - \sum_{l \in \Omega_n^L(+)} F_{l,s,t} \\ & + \sum_{l \in \Omega_n^L(-)} F_{l,s,t} = D_{n,s,t}^{Fore} - D_{n,s,t}^{Cur}, \forall n, \forall t, \forall s \end{aligned} \quad (19)$$

$$0 \leq D_{n,s,t}^{Cur} \leq D_{n,s,t}^{Fore}, \forall n, \forall t, \forall s \quad (20)$$

Equations (19) and (20) describe the power balance and the load shedding constraints of each node.

$$F_{l,s,t} = B_l(\theta_{l(+),s,t} - \theta_{l(-),s,t}), \forall l, \forall t, \forall s \quad (21)$$

$$-F_l^{Max} \leq F_{l,s,t} \leq F_l^{Max} \quad (22)$$

Equations (21) and (22) describe the DC power flow and the power limit of each branch.

$$\sum_{g \in \Omega_c^G} Cap_g - O_{c,s,t} \geq 0; \forall c, \forall s, \forall t \quad (23)$$

$$P_{c,s,t} = \sum_{g \in \Omega_c^G} P_{g,s,t}; \forall c, \forall s, \forall t \quad (24)$$

$$\lambda_c^{Min} O_{c,s,t} \leq \sum_{g \in \Omega_c^G} P_{g,s,t} \leq O_{c,s,t}, \forall c, \forall s, \forall t \quad (25)$$

$$-\alpha_c^{Rd} O_{c,s,t} \leq P_{c,s,t} - P_{c,s,t-1} \leq \alpha_c^{Ru} O_{c,s,t}, \forall c, \forall s, \forall t \quad (26)$$

$$O_{c,s,t} = O_{c,s,t-1} + SU_{c,s,t} - SD_{c,s,t}, \forall c, \forall s, \forall t \quad (27)$$

$$SU_{c,s,t} \leq \sum_{g \in \Omega_c^G} Cap_g - O_{c,s,t-1} - \sum_{\tau=1}^{T_c^{eff}} SD_{c,s,t-\tau}, \forall c, \forall s, \forall t \quad (28)$$

$$SD_{c,s,t} \leq O_{c,s,t-1} - \sum_{\tau=1}^{T_c^{On}} SU_{c,s,t-\tau}, \forall c, \forall s, \forall t \quad (29)$$

Equation (23) limits the maximum capacity of units that can be turned on. Equation (24) limits the output power of the unit cluster to the interval. Equation (25) is the ramp limit. Equations (26)–(29) show the limits of the minimum up and down times. Compared with those of the conventional model, this method uses continuous variables rather than binary variables. In a large system, there are many thermal power units in one node. Therefore, this linear relaxation can greatly reduce the

computation time while maintaining similar performance [26].

$$0 \leq P_{w,s,t} + P_{w,s,t}^{Reg} \leq Cap_w \lambda_{w,s,t}^{Fore}, \forall w, \forall s, \forall t \quad (30)$$

$$0 \leq P_{pv,s,t} + P_{pv,s,t}^{Reg} \leq Cap_{pv} \lambda_{pv,s,t}^{Fore}, \forall pv, \forall s, \forall t \quad (31)$$

Equations (30) and (31) limit the output power of the VRE plants to no larger than the forecasted output.

$$-Cap_{p,es} \leq P_{es,s,t}^{ch} \leq Cap_{p,es}, \forall es, \forall s, \forall t \quad (32)$$

$$-Cap_{p,es} \leq P_{es,s,t}^{dis} \leq Cap_{p,es}, \forall es, \forall s, \forall t \quad (33)$$

$$-Cap_{e,es} \leq E_{es,s,t} \leq Cap_{e,es}, \forall es, \forall s, \forall t \quad (34)$$

$$E_{es,s,t} = E_{es,s,t-1} + P_{es,s,t}^{ch} * \eta_{es} - P_{es,s,t}^{dis} / \eta_{es}, \forall es, \forall s, \forall t \quad (35)$$

$$P_{es,s,t}^{ch} \geq 0, P_{es,s,t}^{dis} \geq 0, \forall es, \forall s, \forall t \quad (36)$$

Equations (32)–(36) are the output limits of energy storage.

$$\begin{aligned} & \sum_c O_{c,s,t} + \sum_w Cap_w \lambda_{w,s,t}^{Fore} + \sum_{pv} \lambda_{pv,s,t}^{Fore} \\ & - \sum_n (D_{n,s,t}^{Fore} - D_{n,s,t}^{Cur}) \leq r_{Load} \sum_n D_{n,t}^{Fore} \\ & + r_{VRE} \left(\sum_w Cap_w \lambda_{w,s,t}^{Fore} + \sum_{pv} Cap_{pv} \lambda_{pv,s,t}^{Fore} \right) \end{aligned} \quad (37)$$

Equation (37) ensures that the online capacity for generation can satisfy the load demand considering the forecast errors for VRE and the load.

B. Frequency Regulation Performance Constraints

As described in Section III, the frequency regulation performance constraints are estimated by a set of rules in the form of a decision tree, which can be written in the form of (14) and (15). The variables in $x_{s,t}$, $\sigma_{s,t}^{load}$, $\sigma_{s,t}^W$ and $\sigma_{s,t}^{PV}$ can be calculated from the historical data. The variables in the operating state $x_{s,t}$, where $R_{c,s,t}^{up}$ and $R_{c,s,t}^{down}$, can be calculated from the variables of the planning problem:

$$R_{c,s,t}^{up} = \alpha_c^{Ru} O_{c,s,t} - P_{c,s,t} + P_{c,s,t-1} \quad (38)$$

$$R_{c,s,t}^{down} = \alpha_c^{Rd} O_{c,s,t} + P_{c,s,t} - P_{c,s,t-1} \quad (39)$$

Equation (40) indicates that the capacity of renewable energy participating in frequency regulation cannot exceed 10% of the total capacity.

$$P_{s,t}^{Reg} \leq \beta \left(\sum_w Cap_w + \sum_{pv} Cap_{pv} \right) \times 10\% \quad (40)$$

where β denotes the proportion of renewable energy that can participate in regulation.

C. Comparison of Traditional and Modified Planning Model

The mathematical formulation of traditional generation expansion planning problem is presented as follows:

$$\min C^{sys} = C^{operation} + C^{investment}$$

TABLE I
PLANNING RESULT OF CASE 1, CASE 2 AND CASE 3

	Capacity (MW)			Total cost ($\times$ billion CNY)	Average Frequency variance (Hz)
	Thermal	Wind	Solar		
Case 1	1.7010 $\times 10^5$	2.9567 $\times 10^5$	9.5335 $\times 10^4$	222.81	0.1073
Case 2	2.9159 $\times 10^5$	3.1890 $\times 10^5$	8.0402 $\times 10^4$	244.16	0.0844
Case 3	2.1930 $\times 10^5$	2.8987 $\times 10^5$	1.0701 $\times 10^5$	239.62	0.0712

$$\text{s.t.} \begin{cases} \text{Investment Constraints (16)–(17)} \\ \text{Unit Commitment Constraints (18)–(37)} \end{cases} \quad (41)$$

Compared to the traditional planning model, the modified model incorporates reserve constraints for linking variables and performance constraints in the form of decision trees. Ultimately, the mathematical formulation is as follows.

$$\begin{aligned} \min C^{sys} &= C^{operation} + C^{investment} \\ \text{s.t.} & \begin{cases} \text{Investment Constraints (16)–(17)} \\ \text{Unit Commitment Constraints (18)–(37)} \\ \text{Reserve Constraint (38)–(40)} \\ \text{Decision Tree Constraint (14)–(15)} \end{cases} \end{aligned} \quad (42)$$

VI. CASE STUDY

A. Test System Setting

A case study on a modified IEEE HRP-38 system is conducted to verify the effectiveness of the proposed method. Some modifications are made to the system. The modified system consists of 38 nodes and 174 branches, 38 coal-fired power units, 8 gas-fired units, 52 photovoltaic power plants, 34 wind farms and three energy storage power stations. Hydropower in the original test is omitted. The parameters of these generators, the transmission line and the load demand can be found in [27].

The planning problem is set to start from scratch, which means that the initial capacity of coal-fired units, gas-fired units, wind power, and photovoltaics is set to zero, and their planned capacity is set as the decision variable. The capacity of the energy storage power station is set at a fixed 200 MW. In order to characterize a high percentage of renewable power system, we require in the constraint that the share of renewable power output needs to be no less than 50% of the total load. The planning of the transmission network is based on Case IV in [27]. The frequency regulation standard is set as $\sigma_f < 0.1$ Hz.

Three cases are compared to verify the effectiveness of the proposed model: Case 1: With no frequency constraints, renewable energy does not participate in frequency regulation. The mathematical formulation of constraints is presented as follows.

Case 2: Frequency constraints are considered, but no renewable energy participates in frequency regulation. Mathematically, $\beta = 0$.

Case 3: Frequency constraints are considered, and up to 50% of renewable energy is able to participate in frequency regulation. Mathematically, $\beta = 0.5$.

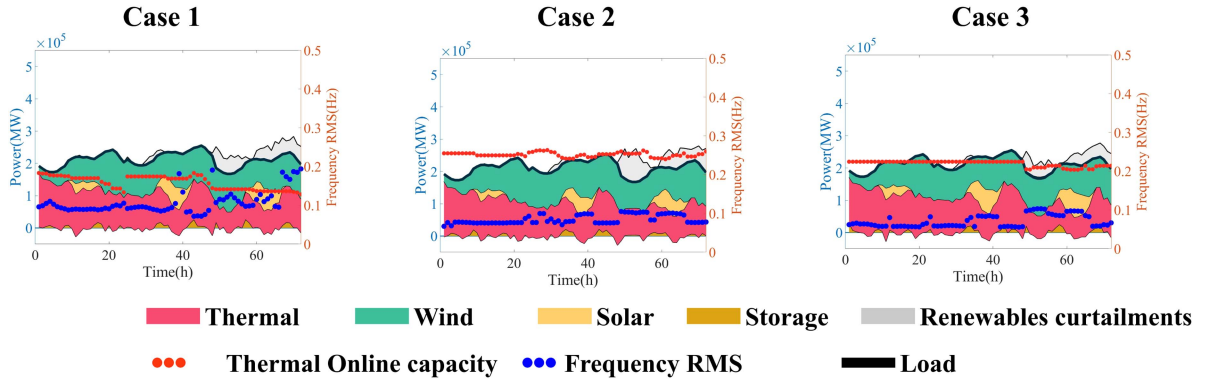


Fig. 8. Power generation results for Case 1, Case 2 and Case 3 (from left to right).

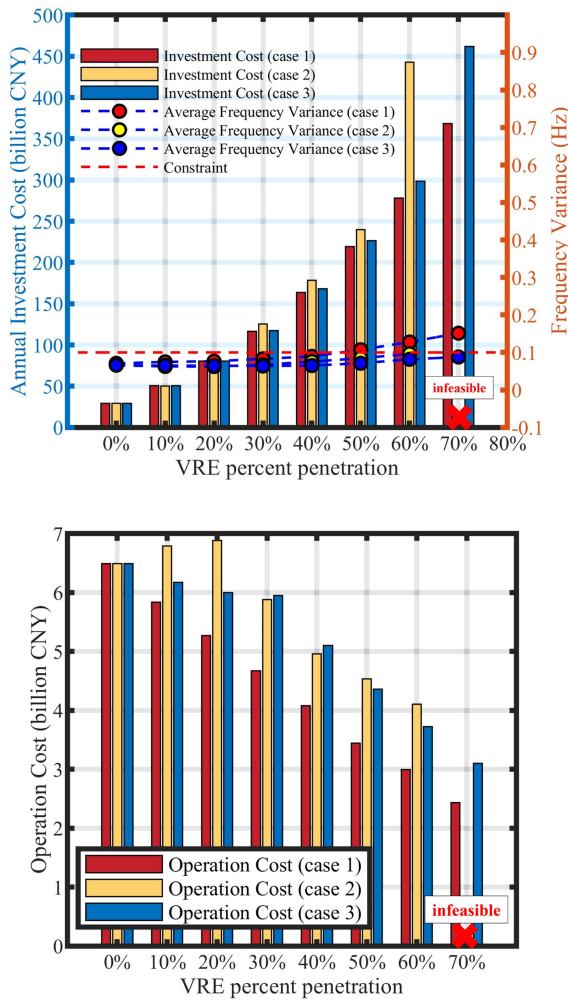


Fig. 9. Operation cost, annual investment cost and average frequency variance in various cases with increasing VRE penetration.

B. Planning Results

The planning results of the three cases are compared in Table I. The results show that the construction capacity of renewable energy is the highest in Case 2 and lowest in Case 1.

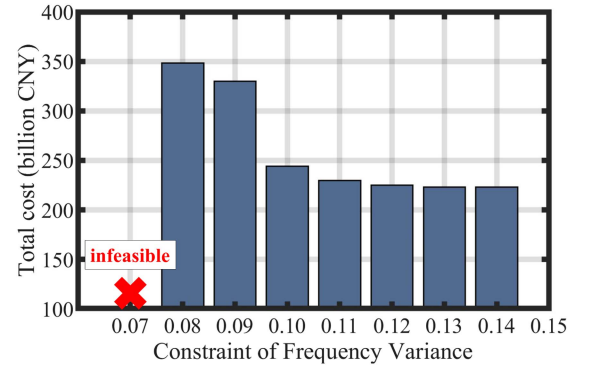


Fig. 10. Total cost in various cases with different constraint of σ_f .

A comparison between Cases 1 and 2 suggests that the consideration of frequency regulation forces the system to need more thermal generators. A comparison between Cases 2 and 3 suggests that the participation of renewables can reduce the requirement for thermal unit online capacity.

Since Case 2 must maintain sufficient online capacity of thermal outputs during the high-output period of renewable energy, to make the penetration rate of renewable energy the same as the other two cases, the renewables have to produce more energy during the rest of the scenario. Therefore, the construction capacity for Case 2 is the highest among the three cases.

The operation results and frequency regulation performance of the three cases in one of the operation scenarios are compared in Fig. 8. This shows that thermal power units maintain a high online capacity in Case 2. Since renewables can provide frequency support in Case 3, the capacity of the online unit is slightly smaller compared with that in Case 2, where renewable energy is reserved during as a backup. Comparatively, in Case 1, some periods cannot meet the system frequency regulation performance standard. Especially from 48 h to 53 h, there is a decrease in system inertia and regulation capacity due to the low online capacity of the thermal power unit and the large output of wind farms. The proposed planning method can efficiently consider the frequency regulation performance requirement in

such periods and can further impact the generation planning result.

C. Sensitivity Analysis

Fig. 9 compares the operation, the annual investment cost and the average frequency variance under increasing VRE targets. When the VRE penetration rate is below 30%, the planning costs for all three sets of cases are almost identical. Compared with that of Case 1, when the VRE penetration rate exceeds 40%, the investment cost increases by 4.5%–59%. Such an increase is only 1.2%–7.3% in Case 3. When VRE targets increase to 70%, the solver fails to find a solution for case 2. This means that the 70% renewable energy target and frequency requirement cannot be achieved simultaneously in this system without the help of renewable energy participation in frequency regulation. The results emphasize the need to consider frequency regulation performance constraints and the effect of frequency support of renewables for high-VRE penetration systems.

The impact of the frequency regulation performance constraint on the results is also taken into consideration to analyze its effect on the system cost. Building upon the settings in Case 2, we adjusted the system's constraint on the performance of frequency regulation. The corresponding results are illustrated in Fig. 10. As the frequency requirement increases, the total cost of the system experiences a significant rise. When the frequency requirement reaches 0.08 Hz, the system cost is 40% higher than that of the original setting. Given that the case setting mandates a renewable energy output share of over 50%, this imposes limitations on the capacity of conventional units. Consequently, it leads to the outcome where the planning problem becomes unsolvable when the frequency requirement reaches 0.07 Hz.

VII. CONCLUSION

The continuous increase in the penetration rate of VRE poses a challenge to the frequency regulation capability of the power system. To confront this issue, we devise a model in this article that delineates the system's frequency regulation performance and accords the constraints to the generation planning model.

A case study based on the HRP-38 system yields the following results: 1) When the VRE penalty surpasses 40%, the frequency constraint makes a considerable discrepancy in the results and has a direct impact on the system's frequency performance. 2) Renewable energy's penetration can exacerbate the system frequency regulation performance, and it can influence the planning result of the system, causing the system's total cost to diminish.

REFERENCES

- [1] National grid, "Frequency quality report." [Online]. Available: <https://www.statnett.no/globalassets/for-aktorer-i-kraftsystemet/utvikling-av-kraftsystemet/nordisk-frekvensstabilitet/frequency-quality-phase-1-v2.0.pdf>
- [2] National grid, "Frequency data for 2016." [Online]. Available: <https://data.nationalgrideso.com/system/system-frequency-data>
- [3] Y. Mi, B. Chen, P. Cai, X. He, R. Liu, and X. Yang, "Frequency control of a wind-diesel system based on hybrid energy storage," *Protection Control Modern Power Syst.*, vol. 7, no. 1, pp. 1–13, 2022.
- [4] G. Magdy, A. Bakeer, and M. Alhasheem, "Superconducting energy storage technology-based synthetic inertia system control to enhance frequency dynamic performance in microgrids with high renewable penetration," *Protection Control Modern Power Syst.*, vol. 6, pp. 1–13, 2021.
- [5] X. Yuan and W. He, "Constraints and solutions to power electronics penetrations in power systems," *iEnergy*, vol. 2, no. 1, pp. 4–5, 2023.
- [6] I. Egido, F. Fernandez-Bernal, P. Centeno, and L. Rouco, "Maximum frequency deviation calculation in small isolated power systems," *IEEE Trans. Power Syst.*, vol. 24, no. 4, pp. 1731–1738, Nov. 2009.
- [7] Z. Chu, U. Markovic, G. Hug, and F. Teng, "Towards optimal system scheduling with synthetic inertia provision from wind turbines," *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 4056–4066, Sep. 2020.
- [8] Z. Zhang, E. Du, F. Teng, N. Zhang, and C. Kang, "Modeling frequency dynamics in unit commitment with a high share of renewable energy," *IEEE Trans. Power Syst.*, vol. 35, no. 6, pp. 4383–4395, Nov. 2020.
- [9] I. Genc, R. Diao, V. Vittal, S. Kolluri, and S. Mandal, "Decision tree-based preventive and corrective control applications for dynamic security enhancement in power systems," *IEEE Trans. Power Syst.*, vol. 25, no. 3, pp. 1611–1619, Aug. 2010.
- [10] Y. Zhang et al., "Encoding frequency constraints in preventive unit commitment using deep learning with region-of-interest active sampling," *IEEE Trans. Power Syst.*, vol. 37, no. 3, pp. 1942–1955, May 2022.
- [11] NERC, "Reliability standards for the bulk electric systems of north America," 2020. [Online]. Available: <https://www.nerc.com/pa/Stand/Reliability%20Standards%20Complete%20Set/RSCCompleteSet.pdf>
- [12] NERC, "Standard bal-001-2—real power balancing control performance," 2020. [Online]. Available: <https://www.nerc.com/pa/Stand/Reliability%20Standards/BAL-001-2.pdf>
- [13] ENTSO-E, "An introduction to network codes & the links between codes," 2020. [Online]. Available: https://www.entsoe.eu/network_codes
- [14] Y. Ba, R. Liu, and W. Li, "Comparison of CPS and its assessment between North America and China," *Dianli Xitong Zidonghua/Automat. Electric Power Syst.*, vol. 36, pp. 63–72, Aug. 2012, doi: 10.3969/j.issn.1000-1026.2012.15.011.
- [15] H. Chavez, R. Baldick, and J. Matevosyan, "The joint adequacy of AGC and primary frequency response in single balancing authority systems," *IEEE Trans. Sustain. Energy*, vol. 6, no. 3, pp. 959–966, Jul. 2015.
- [16] P. Vorobev, D. M. Greenwood, J. H. Bell, J. W. Bialek, P. C. Taylor, and K. Turitsyn, "Deadbands, droop, and inertia impact on power system frequency distribution," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 3098–3108, Jul. 2019.
- [17] F. Galiana, F. Bouffard, J. Arroyo, and J. Restrepo, "Scheduling and pricing of coupled energy and primary, secondary, and tertiary reserves," *Proc. IEEE*, vol. 93, no. 11, pp. 1970–1983, Nov. 2005.
- [18] G. Zhang, E. Ela, and Q. Wang, "Market scheduling and pricing for primary and secondary frequency reserve," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 2914–2924, Jul. 2019.
- [19] S. Pulendran and J. E. Tate, "Capacity scheduling of energy storage and conventional generation for frequency regulation based on CPS1," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 405–414, Jan. 2020.
- [20] L. Liu and Z. Hu, "Data-driven regulation reserve capacity determination based on Bayes theorem," *IEEE Trans. Power Syst.*, vol. 35, no. 2, pp. 1646–1649, Mar. 2020.
- [21] M. E. Haque, S. Paul, and M. R. I. Sheikh, "Grid frequency analysis with the issue of high wind power penetration," in *Proc. Int. Conf. Elect. Inf. Commun. Technol.*, 2014, pp. 1–7.
- [22] P. Bhasaputra, C. Sansilah, and W. Pattaraprakorn, "Impact of solar and wind power development plan in Thailand by considering frequency response of power system," in *Proc. 12th Int. Conf. Elect. Eng./Electron., Comput., Telecommun. Inf. Technol.*, 2015, pp. 1–6.
- [23] F. M. Mele, Á. Ortega, R. Zárate-Miñano, and F. Milano, "Impact of variability, uncertainty and frequency regulation on power system frequency distribution," in *Proc. Power Syst. Comput. Conf.*, 2016, pp. 1–8.
- [24] J. Morsali, K. Zare, and M. T. Hagh, "Appropriate generation rate constraint (GRC) modeling method for reheat thermal units to obtain optimal load frequency controller(LFC)," in *Proc. 5th Conf. Thermal Power Plants*, 2014, pp. 29–34.
- [25] PJM, "Implementation and rationale for pjm's conditionaneutrality regulation signals," 2017. [Online]. Available: <https://pjm.com/-/media/committees-groups/task-forces/rmstf/postings/regulation-market-whitepaper.ashx>
- [26] E. Du, N. Zhang, C. Kang, and Q. Xia, "A high-efficiency network-constrained clustered unit commitment model for power system planning studies," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 2498–2508, Jul. 2019.
- [27] Z. Zhuo et al., "Transmission expansion planning test system for AC/DC hybrid grid with high variable renewable energy penetration," *IEEE Trans. Power Syst.*, vol. 35, no. 4, pp. 2597–2608, Jul. 2020.



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